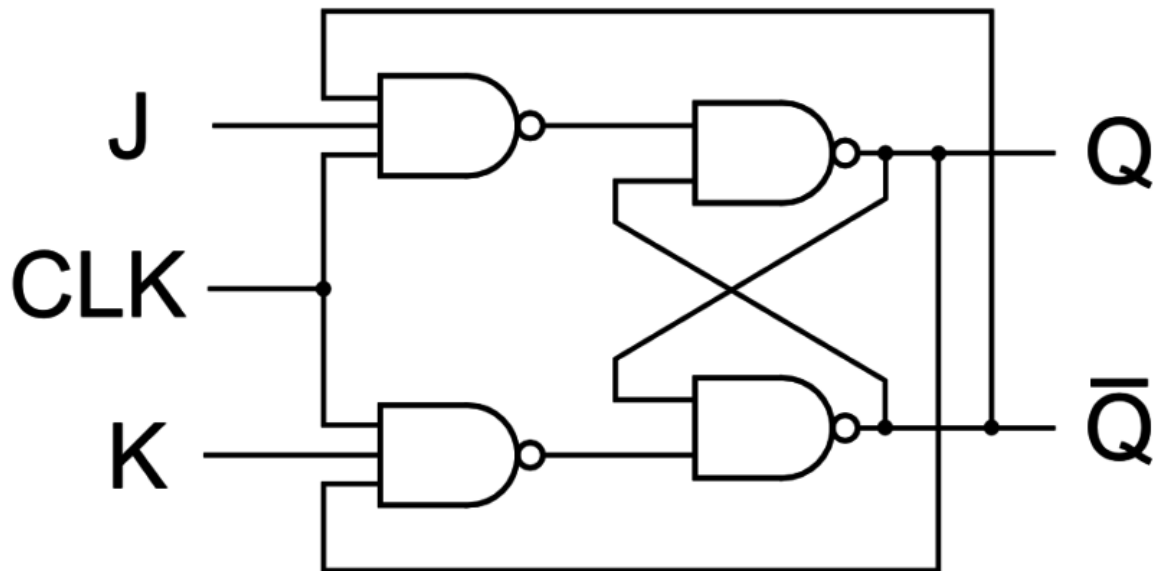


Day 27

JK Flip-Flop (Edge-Triggered) in Verilog

Circuit Diagram :

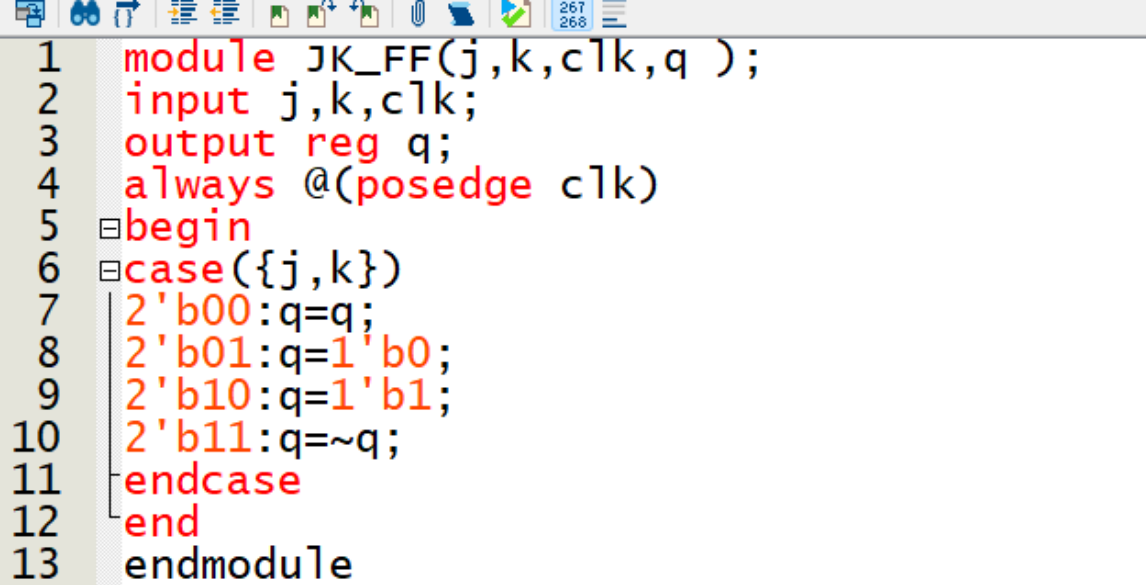


Truth Table

J	K	CLK	Q
0	0	↑	Q_0 (no change)
1	0	↑	1
0	1	↑	0
1	1	↑	$\bar{Q}_0$ (toggles)

Verilog Code

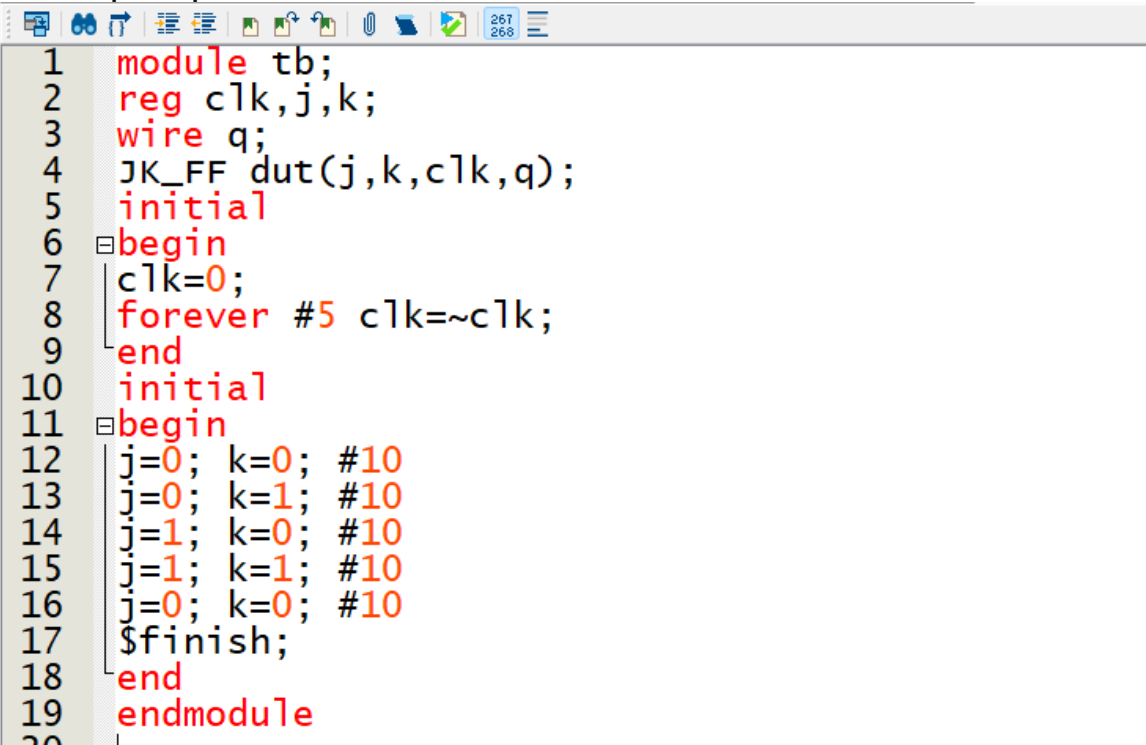
JK-FlipFlop verilog code



```
1  module JK_FF(j,k,clk,q );
2  input j,k,clk;
3  output reg q;
4  always @(posedge clk)
5  begin
6  case({j,k})
7  2'b00:q=q;
8  2'b01:q=1'b0;
9  2'b10:q=1'b1;
10 2'b11:q=~q;
11 endcase
12 end
13 endmodule
```

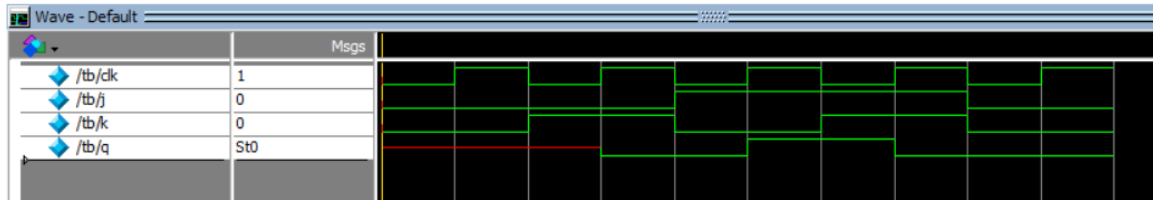
Testbench Code

JK-FlipFlop Tb code

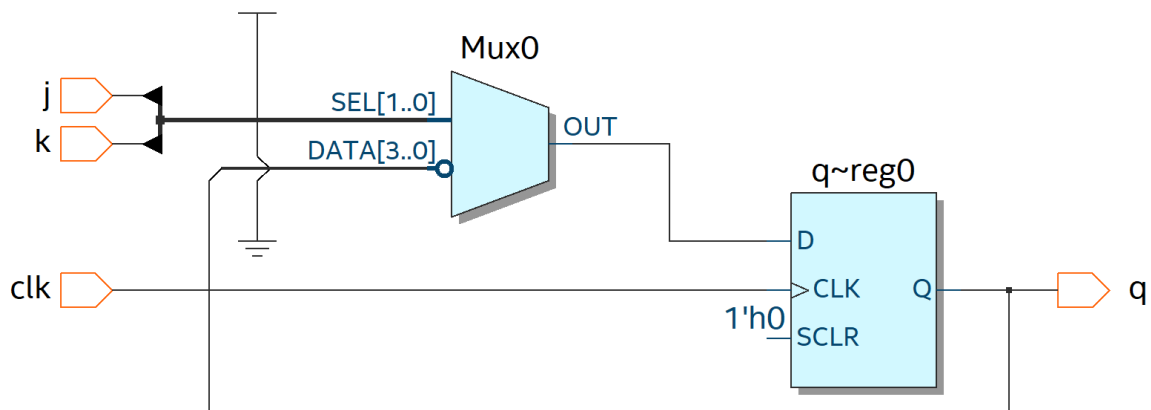


```
1  module tb;
2  reg clk,j,k;
3  wire q;
4  JK_FF dut(j,k,clk,q);
5  initial
6  begin
7  clk=0;
8  forever #5 clk=~clk;
9  end
10 initial
11 begin
12 j=0; k=0; #10
13 j=0; k=1; #10
14 j=1; k=0; #10
15 j=1; k=1; #10
16 j=0; k=0; #10
17 $finish;
18 end
19 endmodule
```

Waveform



Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim